

Oly App Training Logic Documentation

Version: 2.0

Last Updated: February 2026

Purpose: This document defines how the AI generates training blocks, interprets athlete feedback, and adjusts programming. This is the source of truth for all training decisions.

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1. Exercise Library

1.1 Exercise Categories

All exercises are categorized by their programming purpose, not just movement pattern. This enables the AI to select appropriate exercises for each phase of a training session and prescribe targeted accessories based on athlete feedback.

Category hierarchy:

- Competition Lifts (always programmed)
- Competition Lift Variations (conditional)
- Squats (always programmed)
- Pulls (always programmed)

- Squat Accessories (prescribed for weaknesses)
- Pull Accessories (prescribed for weaknesses)
- Overhead Accessories (prescribed for weaknesses)
- Jerk Accessories (prescribed for weaknesses)
- Positional Work (prescribed for technical issues)
- Trunk & Core (prescribed for stability)
- Upper Body GPP (prescribed for general strength)
- Complexes (prescribed for technical coordination)

1.2 Competition Lifts

Purpose: Primary competition movements, always included in programming, percentage-based on own 1RM.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required
Snatch	snatch_1rm	60-100%	1, 2, 3	Barbell, Bumper Plates
Clean & Jerk	clean_jerk_1rm	60-100%	1, 2	Barbell, Bumper Plates

Programming rules:

- Every training session includes 1 competition lift (Snatch OR Clean & Jerk)
- Prescribed first in session order (performed when fresh)
- 4-8 working sets depending on training phase

1.3 Competition Lift Variations

Purpose: Can substitute for or supplement competition lifts. Used to address specific technical weaknesses, develop specific ranges of motion, or modify training stimulus.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Equipment Optional	Addresses Limiting Factors	Addresses Miss
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							Pattern s
Hang Snatch (Knee)	snatch_1rm	65-85 %	1, 2, 3	Barbell, Bumper Plates	-	Speed, Timing	Turnover failures
Hang Snatch (Mid-Thigh)	snatch_1rm	70-90 %	1, 2, 3	Barbell, Bumper Plates	-	Speed, Timing	Turnover failures
Hang Clean (Knee)	clean_jerk_1rm	65-85 %	1, 2, 3	Barbell, Bumper Plates	-	Speed, Timing	Turnover failures
Hang Clean (Mid-Thigh)	clean_jerk_1rm	70-90 %	1, 2, 3	Barbell, Bumper Plates	-	Speed, Timing	Turnover failures
Power Snatch	snatch_1rm	65-85 %	1, 2, 3	Barbell, Bumper Plates	-	Speed	Turnover failures
Power Clean	clean_jerk_1rm	70-90 %	1, 2, 3	Barbell, Bumper Plates	-	Speed	Turnover failures
Block Snatch (Knee)	snatch_1rm	70-90 %	1, 2, 3	Barbell, Bumper Plates, Blocks	-	Position, Speed	Pull phase failures
Block Snatch (Mid-Thigh)	snatch_1rm	75-95 %	1, 2, 3	Barbell, Bumper Plates, Blocks	-	Position, Speed	Pull phase failures
Block Clean (Knee)	clean_jerk_1rm	70-90 %	1, 2, 3	Barbell, Bumper Plates, Blocks	-	Position, Speed	Pull phase failures

Block Clean (Mid-Thigh)	clean_jerk_1rm	75-95 %	1, 2, 3	Barbell, Bumper Plates, Blocks	-	Position, Speed	Pull phase failures
Pause Snatch	snatch_1rm	60-80 %	1, 2, 3	Barbell, Bumper Plates	-	Position	Pull phase failures
Pause Clean	clean_jerk_1rm	65-85 %	1, 2, 3	Barbell, Bumper Plates	-	Position	Pull phase failures
Muscle Snatch	snatch_1rm	50-70 %	3, 4, 5	Barbell, Bumper Plates	-	Timing, Overhead	Turnover failures
Muscle Clean	clean_jerk_1rm	50-70 %	3, 4, 5	Barbell, Bumper Plates	-	Timing	Turnover failures
Tall Snatch	snatch_1rm	40-60 %	3, 4, 5	Barbell, Bumper Plates	-	Speed, Timing	Turnover failures
Tall Clean	clean_jerk_1rm	45-65 %	3, 4, 5	Barbell, Bumper Plates	-	Speed, Timing	Turnover failures

Programming rules:

- Can replace competition lift as primary exercise (e.g., Hang Snatch instead of Snatch)
- Can be added as supplemental work after competition lift (e.g., Snatch + Power Snatch)
- Prescribed when specific limiting factors or miss patterns appear
- Equipment-gated (Block variations only if athlete has blocks)

1.4 Squats

Purpose: Primary lower body strength work, always included in programming, percentage-based on own 1RM.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Addresses Limiting Factors
Front Squat	front_squat_1rm	60-95%	1, 2, 3, 4, 5	Barbell, Bumper Plates, Rack	Legs
Back Squat	back_squat_1rm	65-95%	1, 2, 3, 4, 5	Barbell, Bumper Plates, Rack	Legs
Overhead Squat	snatch_1rm	50-80%	2, 3, 4, 5	Barbell, Bumper Plates, Rack	Overhead, Position

Programming rules:

- Every training session includes 1 squat variation
- Performed after competition lift and pull
- 3-5 working sets depending on training phase
- Front Squat is default for Olympic lifting (more specific to receiving positions)
- Back Squat used for general strength development
- Overhead Squat prescribed when "Overhead" or "Position" limiting factors appear

1.5 Pulls

Purpose: Primary pulling strength work, always included in programming. Develops positional strength and reinforces pull mechanics. Percentage typically 90-110% of corresponding competition lift 1RM.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Equipment Optional	Addresses Limiting Factors	Addresses Miss Patterns
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Snatch Pull	snatch_1rm	90-110 %	2, 3, 4	Barbell, Bumper Plates	Straps	Position, Back	Pull phase failures, Front misses
Clean Pull	clean_jerk_1rm	90-110 %	2, 3, 4	Barbell, Bumper Plates	Straps	Position, Back	Pull phase failures, Front misses
Snatch Deadlift	snatch_1rm	90-120 %	3, 4, 5	Barbell, Bumper Plates	Straps	Position, Back	Pull phase failures
Clean Deadlift	clean_jerk_1rm	95-120 %	3, 4, 5	Barbell, Bumper Plates	Straps	Position, Back	Pull phase failures

Programming rules:

- Every training session includes 1 pull variation
- Performed after competition lift, before squat
- 3-4 working sets
- Straps optional but recommended for heavy pulls (reduces grip fatigue)
- Snatch Pull/Deadlift paired with snatch-focused days
- Clean Pull/Deadlift paired with clean-focused days

1.6 Squat Accessories

Purpose: Prescribed when "Legs" is flagged as limiting factor 5+ times in a week, or for positional/tempo work to improve squat technique.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Addresses Limiting Factors	Prescription Trigger
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Pause Front Squat	front_squat_1 rm	65-85 %	2, 3, 4	Barbell, Bumper Plates, Rack	Legs, Position	"Legs" flagged 5+ times OR "Position" in squat
Pause Back Squat	back_squat_1 rm	70-90 %	2, 3, 4	Barbell, Bumper Plates, Rack	Legs, Position	"Legs" flagged 5+ times OR "Position" in squat
Tempo Front Squat	front_squat_1 rm	60-80 %	3, 4, 5	Barbell, Bumper Plates, Rack	Legs, Position	"Legs" flagged 5+ times OR positional breakdown
Tempo Back Squat	back_squat_1 rm	65-85 %	3, 4, 5	Barbell, Bumper Plates, Rack	Legs	"Legs" flagged 5+ times
Box Squat	back_squat_1 rm	60-80 %	3, 4, 5	Barbell, Bumper Plates, Rack, Box	Legs	"Legs" flagged 5+ times
Bulgarian Split Squat	bodyweight or front_squat_1 rm	0-60%	6, 8, 10 per leg	Barbell or Dumbbells, Bench	Legs	"Legs" flagged 5+ times OR unilateral weakness
Walking Lunges	bodyweight or front_squat_1 rm	0-50%	8, 10, 12 per leg	Barbell or Dumbbells	Legs	"Legs" flagged 5+ times

Programming rules:

- Added to session 4 (after competition lift, pull, squat)
 - 3-4 working sets
 - Pause variations: 2-3 second pause at bottom
 - Tempo variations: 3-4 second eccentric, explosive concentric
 - Not prescribed if athlete already struggling with primary squat volume
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1.7 Pull Accessories

Purpose: Prescribed when "Back" is flagged as limiting factor 5+ times in a week, or to develop posterior chain strength.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Equipment Optional	Addreses Limiting Factors	Contraindications	Prescription Trigger
Romanian Deadlift	back_squat_1rm	50-70 %	6, 8, 10	Barbell, Bumper Plates	Straps	Back	Lower back pain	"Back" flagged 5+ times
Snatch Grip RDL	snatch_1rm	45-65 %	6, 8, 10	Barbell, Bumper Plates	Straps	Back, Position	Lower back pain	"Back" flagged 5+ times AND snatch positioning issues
Clean Grip RDL	clean_jerk_1rm	50-70 %	6, 8, 10	Barbell, Bumper Plates	Straps	Back, Position	Lower back pain	"Back" flagged 5+ times AND clean positioning issues
Good Mornings	back_squat_1rm	40-60 %	6, 8, 10	Barbell, Rack	-	Back	Lower back pain	"Back" flagged

								5+ times
Back Extensions	bodyweight	bodyweight	10, 12, 15	Back Extension Bench or GHD	-	Back	Lower back pain (moderate/severe)	"Back" flagged 5+ times
GHD Back Extensions	bodyweight	bodyweight	8, 10, 12	GHD	-	Back	Lower back pain (moderate/severe)	"Back" flagged 5+ times

Programming rules:

- Added to session 4 (after competition lift, pull, squat)
- 3-4 working sets
- NOT prescribed if athlete reports lower back pain (except Back Extensions at light load)
- Romanian Deadlifts emphasize hamstring stretch, not maximal load

1.8 Overhead Accessories

Purpose: Prescribed when "Overhead" is flagged as limiting factor 5+ times in a week, or when catch phase failures occur in snatch/jerk.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Addreses Limiting Factors	Addreses Miss Patterns	Contraindications	Prescription Trigger
Push Press	clean_jerk_1rm	70-90%	3, 4, 5	Barbell, Bumper	Overhead	Catch phase failures (jerk)	Shoulder pain	"Overhead" flagged 5+ times

				Plates, Rack				OR jerk lockout failures
Strict Press	clean_jerk _1rm	50-7 0%	4, 5, 6	Barbell, Bumpe r Plates, Rack	Overh ead	Catch phase failure s (jerk)	Shoulder pain	"Overhe ad" flagged 5+ times
Behin d-The -Neck Press	snatch_1r m	40-6 0%	5, 6, 8	Barbell, Bumpe r Plates, Rack	Overh ead, Positio n	Catch phase failure s (snatc h)	Shoulder pain	"Overhe ad" flagged 5+ times OR snatch catch instabilit y
Sots Press	snatch_1r m	30-5 0%	5, 6, 8	Barbell, Bumpe r Plates	Overh ead, Positio n	Catch phase failure s (snatc h)	Shoulder pain, ankle mobility limitations	"Overhe ad" flagged 5+ times OR snatch catch instabilit y
Snatc h Balan ce	snatch_1r m	60-8 0%	2, 3, 4	Barbell, Bumpe r Plates	Overh ead, Speed , Positio n	Catch phase failure s (snatc h)	Shoulder pain	"Overhe ad" flagged 5+ times OR 3+ snatch catch failures

Drop Snatch	snatch_1rm	40-60%	3, 4, 5	Barbell, Bumper Plates	Overhead, Speed, Position	Catch phase failures (snatch)	Shoulder pain	"Overhead" flagged 5+ times OR 3+ snatch catch failures
Overhead Holds	snatch_1rm	90-120%	10-30 seconds	Barbell, Bumper Plates, Rack	Overhead	Catch phase failures (snatch)	Shoulder pain, wrist pain	"Overhead" flagged 5+ times OR overhead stability issues
Snatch Press (from bottom)	snatch_1rm	40-60%	4, 5, 6	Barbell, Bumper Plates	Overhead, Position	Catch phase failures (snatch)	Shoulder pain	"Overhead" flagged 5+ times OR snatch catch instability

Programming rules:

- Added to session 4 (after competition lift, pull, squat)
- 2-3 working sets
- NOT prescribed if athlete reports shoulder pain or wrist pain
- Snatch Balance and Drop Snatch are technical drills, not pure strength work
- Overhead Holds performed with 10-30 second isometric holds

1.9 Jerk Accessories

Purpose: Prescribed when jerk-specific weaknesses appear or when lockout failures occur.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Addres ses Limiting Factors	Addres ses Miss Patterns	Prescripti on Trigger
Power Jerk	clean_jerk_1 rm	65-85 %	2, 3, 4	Barbell, Bumper Plates, Rack	Speed, Overhea d	Catch phase failures (jerk)	Jerk lockout issues OR "Speed" flagged in jerk
Split Jerk	clean_jerk_1 rm	70-90 %	1, 2, 3	Barbell, Bumper Plates, Rack	Position, Overhea d	Catch phase failures (jerk)	Jerk receiving position instability
Push Jerk	clean_jerk_1 rm	70-90 %	2, 3, 4	Barbell, Bumper Plates, Rack	Speed, Overhea d	Catch phase failures (jerk)	Jerk timing issues
Jerk Balance	clean_jerk_1 rm	50-70 %	3, 4, 5	Barbell, Bumper Plates, Rack	Position, Speed	Catch phase failures (jerk)	Jerk receiving position instability
Jerk Supports	clean_jerk_1 rm	100-130%	5-10 sec holds	Barbell, Bumper Plates, Rack	Overhea d	Catch phase failures (jerk)	Jerk lockout strength issues

Programming rules:

- Added to session 4 (after competition lift, pull, squat)
- 2-3 working sets
- Jerk Supports are heavy isometric holds in lockout position
- Split Jerk only prescribed if athlete's competition style is split jerk

1.10 Positional Work

Purpose: Prescribed when "Position" is flagged as limiting factor 5+ times in a week, or when 3+ misses occur due to positional breakdown.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Addresses Limiting Factors	Addresses Miss Patterns	Prescription Trigger
Halting Deadlift (Knee)	snatch_1rm or clean_jerk_1rm	60-80 %	3, 4, 5	Barbell, Bumper Plates	Position	Pull phase failures, Front misses	"Position" flagged 5+ times OR 3+ front misses
Halting Deadlift (Mid-T high)	snatch_1rm or clean_jerk_1rm	65-85 %	3, 4, 5	Barbell, Bumper Plates	Position	Pull phase failures, Front misses	"Position" flagged 5+ times OR 3+ front misses
Snatch Segment Pull	snatch_1rm	70-90 %	2+2, 3+3	Barbell, Bumper Plates	Position	Pull phase failures	"Position" flagged 5+ times in snatch
Clean Segment Pull	clean_jerk_1rm	75-95 %	2+2, 3+3	Barbell, Bumper Plates	Position	Pull phase failures	"Position" flagged 5+ times in clean
Snatch High Pull	snatch_1rm	75-95 %	3, 4, 5	Barbell, Bumper Plates	Position, Speed	Pull phase failures	"Position" or "Speed" flagged 5+ times in snatch

Clean High Pull	clean_jerk_1 rm	80-10 0%	3, 4, 5	Barbell, Bumper Plates	Position, Speed	Pull phase failures	"Position" or "Speed" flagged 5+ times in clean
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Programming rules:

- Added to session as pull variation (replaces standard pull) OR as accessory
- 3-4 working sets
- Halting Deadlift: 3-5 second pause at specified height
- Segment Pulls: Pause at floor, then knee/mid-thigh, then finish (2+2 = 2 reps with pauses)
- High Pulls: Emphasize full extension and high elbows, no turnover

1.11 Trunk & Core

Purpose: Prescribed for general trunk stability, injury prevention, or when back position breakdown occurs.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Addresses Limiting Factors	Prescription Trigger
Front Plank	bodyweight	bodyweight	30-60 sec holds	Bodyweight	Back, Position	General trunk stability OR "Position" in squat/pull
Side Plank	bodyweight	bodyweight	30-45 sec holds per side	Bodyweight	Back, Position	General trunk stability OR lateral instability
Ab Wheel	bodyweight	bodyweight	8, 10, 12	Ab Wheel	Back	"Back" flagged OR trunk stability issues

Weighted Sit-ups	light DB/plate	5-20 kg	15, 20, 25	Dumbbells or Plates	Back	"Back" flagged OR trunk strength
Hanging Leg Raises	bodyweight	bodyweight	8, 10, 12	Pull-up Bar	Back	"Back" flagged OR trunk strength
Pallof Press	cable or band	light-moderate	10, 12, 15 per side	Cable Machine or Band	Back, Position	Rotational stability issues

Programming rules:

- Added to end of session (after all barbell work)
- 2-3 working sets
- Often prescribed as circuit (multiple exercises back-to-back)
- NOT prescribed if athlete reports sharp back pain

1.12 Upper Body GPP

Purpose: Prescribed for general upper body strength, pulling strength, and structural balance.

Exercise Name	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Equipment Optional	Addresses Limiting Factors	Prescription Trigger
Pull-ups	bodyweight	bodyweight or weighted	5, 8, 10	Pull-up Bar	Weight Belt	Back, Overhead	General upper body strength OR "Back" flagged

Chin-ups	bodyweight	bodyweight or weighted	6, 8, 10	Pull-up Bar	Weight Belt	Back	General upper body strength OR "Back" flagged
Barbell Rows	back_squat_1rm	40-60%	6, 8, 10	Barbell, Bumper Plates	-	Back	"Back" flagged OR pulling strength
Dumbbell Rows	relative to bodyweight	moderate	8, 10, 12 per side	Dumbbells, Bench	-	Back	"Back" flagged OR unilateral pulling
Dips	bodyweight	bodyweight or weighted	6, 8, 10	Dip Bars	Weight Belt	Overhead	General upper body strength OR "Overhead" in jerk
Face Pulls	cable or band	light-moderate	12, 15, 20	Cable Machine or Band	-	Back, Overhead	Shoulder health OR "Overhead" stability

Programming rules:

- Added to end of session (after all Olympic lifting work)
- 2-3 working sets
- Often prescribed for general physical preparation (GPP) or structural balance
- Pull-ups/Chin-ups can be weighted if athlete completes 10+ bodyweight reps easily

1.13 Complexes

Purpose: Multi-movement combinations prescribed for technical coordination, positional reinforcement, or efficiency in training.

Exercise Name	Movements	Percentage Base	Typical % Range	Typical Reps	Equipment Required	Addresses Limiting Factors	Prescription Trigger
Snatch Pull + Snatch	Snatch Pull, then Snatch	snatch_1rm	65-85 %	1+1, 2+1	Barbell, Bumper Plates	Position, Timing	Positional reinforcement in snatch OR "Position" flagged
Clean Pull + Clean	Clean Pull, then Clean	clean_jerk_1rm	70-90 %	1+1, 2+1	Barbell, Bumper Plates	Position, Timing	Positional reinforcement in clean OR "Position" flagged
Power Snatch + Overhead Squat	Power Snatch, then Overhead Squat	snatch_1rm	60-80 %	1+1, 1+2	Barbell, Bumper Plates	Overhead, Position	"Overhead" in snatch catch OR positional strength
Power Clean + Front Squat	Power Clean, then Front Squat	clean_jerk_1rm	65-85 %	1+1, 1+2	Barbell, Bumper Plates	Legs, Position	"Legs" in clean catch OR positional strength

Clean + Jerk	Clean, then Jerk	clean_jerk_1rm	70-95 %	1+1	Barbell, Bumper Plates	Timing	Competition practice OR jerk timing
Snatch + Overhead Squat	Snatch, then Overhead Squat	snatch_1rm	65-85 %	1+1, 1+2	Barbell, Bumper Plates	Overhead, Position	"Overhead" in snatch catch OR positional strength
Halting Clean Deadlift + Clean Pull + Clean	Halting Clean DL, Clean Pull, Clean	clean_jerk_1rm	60-80 %	1+1+1	Barbell, Bumper Plates	Position, Timing	Positional breakdown in clean OR technical coordination

Programming rules:

- Can replace primary competition lift OR be added as supplemental work
- Performed as single set (all movements completed without dropping bar)
- Rest 2-3 minutes between complex sets
- Notation: "1+1" means 1 rep of first movement, 1 rep of second movement
- Prescribed when technical coordination between movements needs reinforcement

1.14 Exercise Database Schema

Each exercise in the database includes the following metadata:

json

```
{
  "id": "uuid",
  "name": "Snatch",
  "category": "competition_lifts",
  "tier": "core",
  "percentage_base": "snatch_1rm",
  "percentage_range": [60, 100],
```

```

"typical_reps": [1, 2, 3],
"equipment_required": ["barbell", "bumper_plates"],
"equipment_optional": [],
"addresses_limiting_factors": [],
"addresses_miss_patterns": [],
"contraindications": [],
"progression_variants": ["hang_snatch", "power_snatch", "block_snatch"],
"prescription_trigger": null
}

```

Field definitions:

Field	Type	Purpose
id	UUID	Unique identifier
name	String	Canonical exercise name (used in UI, logging, programming)
category	Enum	Programming category (see Section 1.1)
tier	Enum	"core" (always programmed), "variation" (conditional), "accessory" (prescribed)
percentage_base	String	Which 1RM to calculate from (snatch_1rm, clean_jerk_1rm, front_squat_1rm, back_squat_1rm)
percentage_range	Array[Int]	Typical training intensity range [min%, max%]
typical_reps	Array[Int]	Common rep schemes [1, 2, 3] or [3, 4, 5]
equipment_required	Array[String]	Mandatory equipment (filters exercises based on athlete's available equipment)
equipment_optional	Array[String]	Nice-to-have equipment (e.g., straps for pulls)
addresses_limiting_factors	Array[String]	Which limiting factors this exercise targets (legs, back, overhead, speed, position, timing)

<code>addresses_miss_patterns</code>	Array[String]	Which miss patterns this exercise corrects (front_miss, behind_miss, pull_failure, turnover_failure, catch_failure)
<code>contraindications</code>	Array[String]	When NOT to prescribe (shoulder_pain, wrist_pain, lower_back_pain, knee_pain)
<code>progression_variants</code>	Array[String]	Related exercises for scaling or progression
<code>prescription_trigger</code>	String	Condition that triggers prescription (e.g., "Legs flagged 5+ times")

2. Core Programming Principles

2.1 Percentage-Based Loading

All competition lifts (snatch and clean & jerk), their variations, squats, and pulls are prescribed as a percentage of the athlete's current 1RM. Percentages are adjusted based on training phase and athlete feedback.

Standard Percentage Ranges by Goal:

Goal	Percentage Range	Use Case
Technique/Skill work	60-75%	Accumulation phase, technical refinement
Strength-Speed	75-85%	Accumulation phase, building work capacity
Speed-Strength	85-92%	Intensification phase, power development
Max Effort	92-100%	Realization phase, competition prep

2.2 Volume Prescription

Sets × Reps are prescribed per exercise. Volume is adjusted by training phase (see Section 5: Training Block Structure).

Rep Schemes:

Rep Range	Percentage Range	Use Case
Singles (1 rep)	85%+	Max effort work, Intensification/Realization phases
Doubles/Triples (2-3 reps)	75-85%	Standard training, all phases
Higher reps (4-6)	60-75%	Accessory work, Accumulation phase

2.3 Percentage Calculation for Exercise Variations

Different exercise categories use different 1RM bases:

Exercise Category	Percentage Base	Calculation Method
Snatch variations	Snatch 1RM	Direct percentage
Clean variations	Clean & Jerk 1RM	Use Clean 1RM component (typically 85% of C&J)
Snatch Pulls/Deadlifts	Snatch 1RM	90-110% (heavier than competition lift)
Clean Pulls/Deadlifts	Clean & Jerk 1RM	90-120% (heavier than competition lift)
Front Squat	Front Squat 1RM	Direct percentage
Back Squat	Back Squat 1RM	Direct percentage
Overhead Squat	Snatch 1RM	50-80% (lighter than snatch)
Squat accessories	Parent squat 1RM	60-90% (lighter than parent)

Pull accessories	Back Squat 1RM or relative	40-70%
Overhead accessories	Snatch or C&J 1RM	30-90% depending on exercise

MVP Simplification:

For MVP, onboarding collects 4 1RMs: Snatch, Clean & Jerk, Front Squat, Back Squat. All percentage calculations derive from these.

3. Athlete Feedback System

3.1 Required Data Per Set

Athletes log every working set with the following data:

Data Point	Type	Source
Prescribed weight	kg, % of 1RM	Auto-calculated by system
Actual weight lifted	kg	Athlete input
Reps completed	Number	Athlete input
Make/Miss	Binary	Athlete input
Video upload	File	Optional — User Videos Archive

3.2 Lift Analysis Data (Collected Per Set)

A. Bar Speed

Rating Scale: Poor → Acceptable → Good → Excellent

Rating Definitions:

- **Poor:** Too slow, visible deceleration
- **Acceptable:** Slightly slow, but completed

- **Good:** Fast enough, clean movement
- **Excellent:** Very explosive, effortless

AI Use Case:

Track bar speed trends. If Bar Speed is Poor on 50%+ of sets for 2 consecutive weeks → reduce intensity OR reduce volume.

B. Position Quality

Rating Scale: Poor → Acceptable → Good → Excellent

Rating Definitions:

- **Poor:** Lost positions, significant breakdown
- **Acceptable:** Minor breakdown, still completed
- **Good:** Positions held, minor technical deviation
- **Excellent:** Positions locked, textbook execution

AI Use Case:

Track position quality trends. If Position Quality is Poor on 50%+ of sets → reduce intensity AND prescribe positional work.

C. Primary Limiting Factor (Multi-Select)

Athlete selects what limited the lift:

Factor	Description
Legs	Insufficient leg drive, weak out of the hole
Back	Weak trunk, back rounding, unable to maintain position
Speed	Bar moved too slowly
Overhead	Weak lockout, unstable catch in snatch/jerk
Position	Positional breakdown led to failure
Timing	Mistimed pull, rushed transition, coordination issue

AI Use Case:

Track patterns over time. If same limiting factor appears 5+ times in a week → prescribe targeted accessory work from corresponding category (Section 1.6-1.12).

3.3 Miss Tracking (Only If Lift Missed)

Was It a Miss?

- **Yes/No toggle**
- If Yes → collect detailed miss data

Where Did It Fail? (Phase of Lift)

Phase	Description
Pull	Failed during first pull or transition
Turnover	Failed getting under the bar
Catch / Overhead	Failed in the receiving position or lockout

Missed Where? (Direction)

Direction	Description
In front	Bar crashed forward
Behind	Bar went backward
Left	Bar drifted left
Right	Bar drifted right

AI Use Cases:

- **3+ "In front" misses in a week** → Bar is looping, prescribe positional/pull work (Halting Deadlifts, Snatch/Clean Pulls, Segment Pulls)
- **3+ "Behind" misses** → Over-pulling or weak catch, prescribe overhead work (Snatch Balance, Tall Snatches/Cleans)

- **Lateral misses (Left/Right)** → Setup or imbalance issue, flag for coach review (may indicate injury or mobility limitation)
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3.4 Pain & Discomfort Tracking

Any Pain or Discomfort?

- **Yes/No toggle**
- If Yes → collect pain details

Pain Level

Level	Definition
None	Normal soreness
Minor	Sore but manageable
Moderate	Affecting my technique
Sharp	Can't lift safely

Where? (Body Location — Multi-Select)

Available options: Shoulder, Elbow, Wrist, Back, Hip, Knee, Ankle, Other

AI Use Cases:

- **Sharp pain** → Flag session, suggest rest or medical review, remove exercise from programming
 - **Moderate pain 2+ sessions** → Reduce volume by 30% OR intensity by 5% in affected movement
 - **Consistent pain location** → Adjust exercise selection (e.g., shoulder pain → reduce overhead volume, add prehab), cross-reference with contraindications in Exercise Library
-

4. Auto-Regulation Rules (MVP)

4.1 Success-Based Adjustments (Per Lift Type)

Track make/miss rate per lift over a training week (Mon-Sun).

Decision Matrix:

Make Rate	Bar Speed	Position Quality	Action
≥90% (9-10/10)	Good/Excellent on 80%+	Good/Excellent on 80%+	Increase % by 2.5% next week
70-89% (7-8/10)	Any	Any	Maintain current % next week
<70% (≤6/10)	Any	Any	Reduce % by 5% next week

Note: These adjustments are modulated by Coaching Approach setting (see Section 8.2, Step 6).

4.2 Quality-Based Overrides (Takes Priority Over Make Rate)

Poor Bar Speed on 50%+ of sets (even if lifts made)

→ **Action:** Reduce intensity by 5% OR reduce volume by 20% (AI decides based on training phase and Coaching Approach)

Poor Position Quality on 50%+ of sets

→ **Action:** Reduce intensity by 5% AND prescribe positional work next session (from Positional Work category, Section 1.10)

Same Primary Limiting Factor on 5+ sets in a week

→ **Action:** Add targeted accessory work from corresponding category:

Limiting Factor	Accessory Category to Prescribe From
Legs	Squat Accessories (Section 1.6)
Back	Pull Accessories (Section 1.7)
Overhead	Overhead Accessories (Section 1.8)
Speed	Competition Lift Variations (Power/Hang variations, Section 1.3)

Position	Positional Work (Section 1.10)
Timing	Complexes (Section 1.13) OR Competition Lift Variations (Hang/Tall, Section 1.3)

Selection logic:

1. Filter exercises from appropriate category
2. Check equipment availability (athlete must have required equipment)
3. Check contraindications (don't prescribe if athlete has reported relevant pain)
4. Select 1-2 exercises, 3-4 sets each

4.3 Miss Pattern-Based Adjustments

3+ Misses in Same Direction (Front/Behind) in One Week

Adjust technique focus next week:

Miss Direction	Prescription
Front misses	Snatch/Clean Pulls, Halting Deadlifts, Segment Pulls (from Pulls or Positional Work categories)
Behind misses	Hang variations (Hang Snatch/Clean), Tall Snatches/Cleans, Overhead accessories (Snatch Balance, Drop Snatch)

Selection logic:

1. Identify miss direction pattern
2. Prescribe 1-2 exercises from corresponding category as primary or supplemental work
3. Reduce primary competition lift volume by 20% to accommodate corrective work

3+ Misses in Same Phase (Pull/Turnover/Catch)

Adjust exercise selection next week:

Fail Phase	Prescription
-------------------	---------------------

Pull failures	Pulls (Snatch/Clean Pull, Segment Pulls) OR Positional Work (Halting Deadlifts)
Turnover failures	Competition Lift Variations (Muscle Snatch/Clean, Tall Snatch/Clean, Power variations)
Catch failures	Overhead Accessories (Snatch Balance, Drop Snatch, Overhead Squats) OR Jerk Accessories (Jerk Balance, Power Jerk)

Selection logic:

1. Identify fail phase pattern
 2. Replace or supplement primary competition lift with corresponding variation
 3. Example: If 3+ snatch turnover failures → replace Snatch with Hang Snatch (Mid-Thigh) for 1-2 weeks
-

4.4 Pain-Based Overrides (Highest Priority)

Sharp Pain Reported

- **Immediate flag:** "Stop session, seek medical review"
- **Action:** Remove exercise from next week's programming, prescribe alternative if possible
- **Example:** Sharp shoulder pain in snatch → remove all overhead movements, focus on squat and pull work

Moderate Pain 2+ Sessions in Same Location

- **Action:** Reduce volume by 30% in affected movements OR reduce intensity by 5%
- **Action:** Add prehab/mobility work for affected area (if available in exercise library)
- **Example:** Moderate knee pain 2+ sessions → reduce squat volume from 5 sets to 3 sets, add ankle mobility work

Minor Pain 3+ Sessions

- **Action:** Flag for coach review
- **Action:** Consider exercise substitution using Exercise Library contraindications
- **Example:** Minor low back pain 3+ sessions in Back Squat → substitute Front Squat (more upright torso, less low back stress)

Equipment for pain management:

Cross-reference Exercise Library **contraindications** field. If athlete reports pain in location X, filter out exercises with X in contraindications.

5. Training Block Structure (MVP)

5.1 Block Types (Periodization Phases)

Accumulation Block (4-6 weeks)

Goal: Build work capacity, reinforce technique, positional strength

Attribute	Value
Volume	High (total sets/reps)
Intensity	Moderate (70-85% of 1RM)
Rep schemes	3-5 reps per set
Focus	Positional consistency under moderate load
Exercise selection	Competition Lifts + Pulls + Squats (from Sections 1.2, 1.4, 1.5)

Intensification Block (3-4 weeks)

Goal: Increase power output, speed-strength

Attribute	Value
Volume	Moderate (fewer sets/reps)
Intensity	High (80-92% of 1RM)
Rep schemes	1-3 reps per set
Focus	Bar speed, explosive power
Exercise selection	Competition Lifts + Competition Lift Variations + Pulls + Squats

Realization Block (1-2 weeks)

Goal: Peak for competition or testing

Attribute	Value
Volume	Low (minimal sets/ reps)
Intensity	Very High (85-100% of 1RM)
Rep schemes	Singles (1 rep), max effort
Focus	Competition readiness, PRs
Exercise selection	Competition Lifts only + minimal Squats

MVP Simplification:

For MVP, all athletes run continuous Accumulation blocks. Intensification and Realization phases are added in Production after core auto-regulation is validated.

5.2 Deload Weeks

Triggered by:

- Every 4-6 weeks (programmed)
- Moderate pain 2+ sessions
- Make rate <60% for 2 consecutive weeks
- Poor Bar Speed + Poor Position Quality on 50%+ of sets for 2 weeks

Deload Structure:

- Reduce volume by 40-50%
- Reduce intensity by 10-15%
- Maintain movement patterns (still do comp lifts, just lighter/less)

Example Deload Week:

Standard Week	Deload Week
Snatch 5×3 @ 80%	Snatch 3×2 @ 70%

Snatch Pull 4×3 @
95%

Snatch Pull 2×3 @
85%

Front Squat 5×3 @
80%

Front Squat 3×3 @
70%

6. Session Structure (MVP)

6.1 Standard Training Session Order

Exercise Order (Fixed):

1. **Competition Lift** (from Section 1.2) OR **Competition Lift Variation** (from Section 1.3)
 - 4-8 working sets
 - Percentage-based on training phase
 - Examples: Snatch, Clean & Jerk, Hang Snatch, Power Clean
2. **Pull Variation** (from Section 1.5)
 - 3-4 working sets
 - Percentage-based (often 90-110% of comp lift 1RM)
 - Examples: Snatch Pull, Clean Pull, Snatch Deadlift
3. **Squat Variation** (from Section 1.4)
 - 3-5 working sets
 - Percentage-based
 - Examples: Front Squat, Back Squat, Overhead Squat
4. **Accessory Work** (from Sections 1.6-1.12) (*Optional, 1-2 exercises*)
 - Prescribed based on limiting factors, miss patterns, or pain management
 - 2-4 working sets
 - Examples: Pause Squats, Romanian Deadlifts, Push Press, Planks

Rationale:

Most technical and demanding lifts performed first when fresh. Strength work follows. Accessories target weaknesses last.

6.2 Weekly Training Frequency (MVP)

Standard Structure:

Training Days	Weekly Distribution
3 days	Day 1: Snatch focus, Day 2: C&J focus, Day 3: Combined OR Squat emphasis
4 days	Heavy / Light / Medium / Heavy (alternating intensities)

Rules:

- At least 1 rest day between heavy sessions
- Competition lifts rotated (Snatch-focused days alternate with C&J-focused days)
- If athlete trains 3 days, structure is: Mon (Snatch), Wed (C&J), Fri (Snatch OR Combined)
- If athlete trains 4 days, structure is: Mon (Heavy Snatch), Tue (Light C&J), Thu (Medium Snatch), Fri (Heavy C&J)

Exercise selection by day:

Day Type	Competition Lift	Pull	Squat	Accessories
Snatch-focused	Snatch OR Snatch variation	Snatch Pull OR Snatch Deadlift	Front Squat OR Overhead Squat	Based on limiting factors
C&J-focused	Clean & Jerk OR Clean variation	Clean Pull OR Clean Deadlift	Front Squat OR Back Squat	Based on limiting factors
Combined	Snatch + Clean & Jerk (both lighter)	Snatch Pull OR Clean Pull	Front Squat	Minimal accessories

7. AI Decision Tree (How AI Uses This Data)

Step 1: Check for Red Flags (Highest Priority)

Red Flag	Action

Sharp pain?	Stop programming affected movement, flag for review, filter exercises using contraindications field in Exercise Library
Moderate pain 2+ sessions?	Reduce volume by 30% OR intensity by 5%, cross-reference Exercise Library for alternatives
Make rate <60% for 2 weeks?	Trigger deload (Section 5.2)

Step 2: Analyze Lift Quality (Per Lift Type)

Quality Issue	Action
Bar Speed Poor on 50%+ sets?	Reduce intensity by 5% OR reduce volume by 20% (decision modulated by Coaching Approach)
Position Quality Poor on 50%+ sets?	Reduce intensity by 5%, prescribe from Positional Work category (Section 1.10)
Same limiting factor 5+ times?	Add targeted accessory from corresponding category (Section 4.2 mapping)

Example:

If "Legs" flagged 5+ times this week:

1. Query Exercise Library: **category == "squat_accessories"**
 2. Filter by **equipment_required** (athlete must have equipment)
 3. Filter by **contraindications** (don't prescribe if athlete has knee pain)
 4. Select 1-2 exercises: Pause Front Squat, Tempo Back Squat
 5. Add to next week's programming: 3-4 sets each
-

Step 3: Analyze Miss Patterns

Pattern	Action
3+ misses same direction (front/behind)?	Adjust technique focus using Section 4.3 mapping

3+ misses same phase
(pull/turnover/catch)?

Adjust exercise selection using Section 4.3
mapping

Example:

If 3+ snatch misses "In Front" this week:

1. Query Exercise Library: `category == "pulls" OR category == "positional_work"`
 2. Filter: `addresses_miss_patterns` contains "front_miss"
 3. Select: Halting Deadlift (Knee), Snatch Segment Pull
 4. Prescribe: Replace standard Snatch Pull with Halting Deadlift next week
-

Step 4: Adjust Programming for Next Week

Condition	Action
Make rate $\geq 90\%$ AND quality good	Increase % by 2.5% (modulated by Coaching Approach)
Make rate 70-89%	Maintain current %
Make rate $< 70\%$	Decrease % by 5% (modulated by Coaching Approach)

Step 5: Generate Next Week's Block

Process:

1. Apply adjusted percentages from Step 4
2. Add targeted accessories from Step 2 (limiting factors)
3. Adjust exercise selection from Step 3 (miss patterns)
4. Maintain block structure (Accumulation phase for MVP)
5. Apply Coaching Approach and Training Emphasis preferences (Section 8.2, Step 6)
6. Validate equipment requirements (filter exercises athlete doesn't have equipment for)
7. Validate contraindications (filter exercises that conflict with reported pain)

Example week generation:

json

```
{
  "week": 2,
  "phase": "accumulation",
  "adjustments_applied": [
    "Increased snatch % from 75% to 77.5% (make rate 95%)",
    "Added Pause Front Squat (Legs flagged 6 times)",
    "Replaced Snatch Pull with Halting Deadlift (3 front misses)"
  ],
  "sessions": [
    {
      "day": 1,
      "focus": "snatch",
      "exercises": [
        {
          "name": "Snatch",
          "category": "competition_lifts",
          "sets": 5,
          "reps": 3,
          "intensity": 77.5,
          "weight_kg": 85.25
        },
        {
          "name": "Halting Deadlift (Knee)",
          "category": "positional_work",
          "sets": 4,
          "reps": 4,
          "intensity": 70,
          "weight_kg": 77
        },
        {
          "name": "Front Squat",
          "category": "squats",
          "sets": 4,
          "reps": 4,
          "intensity": 77.5,
          "weight_kg": 124
        },
        {
          "name": "Pause Front Squat",
```

```
"category": "squat_accessories",
"sets": 3,
"reps": 3,
"intensity": 70,
"weight_kg": 112
}
]
}
]
```

8. New User Onboarding

8.1 Initial Phase Assignment

Rule: All new users start in Accumulation Phase (4-6 weeks)

Rationale:

- Safest entry point—moderate intensity (70-85%), higher volume
 - Diagnostic period: reveals true technical ability, work capacity, and movement patterns
 - Their stated 1RMs may be inaccurate (ego, outdated, tested when fresh)
 - First 2-3 weeks provide baseline data for auto-regulation
 - Less intimidating than heavy singles for new users
-

8.2 Onboarding Data Collection

Step 1: Collect Current 1RMs

Required inputs:

- Snatch: ____ kg
 - Clean & Jerk: ____ kg
 - Front Squat: ____ kg
 - Back Squat: ____ kg
-

Step 2: Training Considerations

"Used to modify training when needed."

Data collected:

Field	Type	Options
Training Limitations	Toggle	Yes / No
Affected Areas	Multi-select	Lower back, Knees, Shoulders, Wrists, Hips, Ankles
Current Impact	Single-select	Mild / Moderate / High
When Does It Show Up	Multi-select	Overhead position, Snatch catch, Jerk lockout, High OH volume, When fatigued, Next-day soreness

How this affects programming:

Cross-referenced with Exercise Library **contraindications** field. If athlete selects "Shoulders" with "Moderate" impact, AI filters out exercises with **contraindications: ["shoulder_pain"]** from prescription logic.

Step 3: Availability

"Used to structure your weekly training."

Data collected:

Field	Type	Options
Training Days Per Week	Single-select	2 / 3 / 4 / 5 / 6
Session Duration	Single-select	45 min / 60 min / 75 min / 90+ min
Preferred Rest Days	Multi-select	Mon, Tue, Wed, Thu, Fri, Sat, Sun

How this affects programming:

- Training days determines weekly distribution (Section 6.2)
 - Session duration determines exercise count (45min = 3 exercises, 60min = 4 exercises, 75min = 4 + accessories, 90min = 4 + extended accessories)
 - Preferred rest days used to schedule training sessions
-

Step 4: Equipment

"Used to tailor exercise selection and loading."

Essential Equipment (auto-selected, cannot uncheck):

- Barbell + Bumper Plates
- Squat Rack

Optional Equipment (checkboxes):

- Lifting Blocks
- Pull-up Bar
- Dumbbells & Kettlebells
- GHD Machine

How this affects programming:

Filters Exercise Library by `equipment_required` field. If athlete doesn't have Lifting Blocks, all exercises with `equipment_required: ["blocks"]` are excluded from programming (e.g., Block Snatch, Block Clean).

Step 5: Training Preference

"Select how you prefer training to be emphasized."

Options:

Preference	Tag	Description
High Intensity	HIGH LOAD	Frequent heavy singles and doubles to prioritize neural output

Balanced	PROGRESSIV E	Structured progression with balanced volume and intensity (<i>Default</i>)
Higher Volume	REPETITION	Higher repetition work to build technical and work capacity
Adaptive	-	Automatically adjusts based on readiness and recent performance

How this affects programming:

Preference	Starting Intensity	Rep Schemes	Accumulation Duration
High Intensity	75-80%	Doubles	4 weeks
Balanced (<i>Default</i>)	70-85%	3-5 reps	4-6 weeks
Higher Volume	70-75%	3-5 reps	5-6 weeks
Adaptive	70-85%	3-5 reps	Variable (based on make rate + quality)

Step 6: Coaching Approach

"How your coach behaves, manages risk, and adjusts programming."

Coaching Approach:

"How much your coach pushes training intensity and manages risk."

Option	Behavior
Conservative	Prioritizes safety and technique, slower progression, larger safety margins
Normal	Balanced approach to progression and risk management (<i>Default</i>)
Aggressive	Pushes intensity more aggressively, faster progression, smaller safety margins

How this affects auto-regulation (Section 4.1-4.4):

Coaching Approach	Increase %	Decrease %	Deload Trigger	Position Quality Override
Conservative	+2%	-7.5%	Make rate <65%	Immediate load reduction on Poor
Normal	+2.5%	-5%	Make rate <60%	Immediate load reduction on Poor
Aggressive	+3%	-3%	Make rate <55%	Allow 2-3 Poor ratings before reduction

Training Emphasis:

"Determines whether training prioritizes technical quality or loading progression."

Option	Behavior
Technique-first	Prioritizes positional quality over load increases (<i>Default</i>)
Strength-first	Prioritizes load progression, accepts minor technique compromises

How this affects auto-regulation:

Training Emphasis	Position Quality Threshold	Load Increase Requirement	Accessory Priority
Technique-first	Poor position = immediate load reduction	Bar speed AND position must be Good/Excellent	Positional Work (Section 1.10)
Strength-first	2-3 Poor ratings before load reduction	Bar speed OR position can be Acceptable	Squat/Pull Accessories (Sections 1.6, 1.7)

Feedback Style:

"How direct your coach should be when giving cues and feedback."

Option	Behavior
--------	----------

Supportive	Encouraging, positive reinforcement, gentle corrections
Calm	Neutral, matter-of-fact, minimal emotional tone
Balanced	Mix of encouragement and directness
Direct	Straightforward, no sugar-coating, efficient communication

How this affects system messages:

Controls tone of all AI-generated feedback, session summaries, and programming explanations. Does NOT affect programming logic, only communication style.

Context: Training Decisions:

"When enabled, your coach explains why specific exercises, variations, and loading schemes are chosen."

Option	Behavior
Toggle (On/Off)	When enabled, AI provides rationale for programming decisions

Example with Context enabled:

"This week's snatch percentage increased to 77.5% because you made all lifts last week with Good bar speed and Excellent position quality. Your limiting factor data shows consistent 'Legs' flags, so we're adding Pause Front Squats to build strength in that area."

Example with Context disabled:

"Week 2: Snatch 5×3 @ 77.5%, Snatch Pull 4×3 @ 95%, Front Squat 4×4 @ 77.5%, Pause Front Squat 3×3 @ 70%"

Step 7: System Message to User

"Your first training block will be a 4-week Accumulation Phase. This helps us understand your movement quality and work capacity. You'll train at 70-85% intensity with higher volume. Focus on solid technique and consistent feedback—we'll adjust from there."

8.3 Week 1 Programming (Conservative Start)

Base Week 1 Structure

Before adjustments:

Exercise	Category	Sets × Reps	Intensity
Snatch	Competition Lifts	4 × 3	70-75%
Snatch Pull	Pulls	3 × 3	90% (of Snatch 1RM)
Front Squat	Squats	4 × 3	75%

Rationale for conservative start:

- Uses only core exercises from Sections 1.2, 1.4, 1.5
- No variations, no accessories (diagnostic week)
- Percentages at lower end of Accumulation range
- 3-rep sets (moderate volume)

Adjustments Based on Onboarding

Modification rules:

Condition	Adjustment
Training Limitations: Moderate/High	Reduce to 3 sets per exercise
Session Duration: 45 min	Remove Snatch Pull (only Competition Lift + Squat)
Training Days: 2 per week	Combine into full-body sessions (Snatch + Clean & Jerk + Squat)

Training Preference: High Intensity	Start at 75%, use doubles instead of triples
Training Preference: Higher Volume	Start at 70%, use 4-5 reps
Coaching Approach: Conservative	Start at 70% (lower end)
Coaching Approach: Aggressive	Start at 75% (upper end)
Training Emphasis: Strength-first	Add Back Squat volume (4×3 @ 75%)
Equipment: No Blocks	(No adjustment, blocks not used in Week 1)

8.4 First 2 Weeks: Assessment Period

Week 1 Data Collection

AI monitors:

- Make rate (should be 95-100% at these percentages)
 - Bar speed (should be Good/Excellent consistently)
 - Position quality (can they hold positions under volume?)
 - Pain reports (red flag if pain appears at 70-75%)
 - Limiting factors (track patterns)
-

Week 2 Adjustments Based on Week 1 Performance

Scenario A: Crushed Week 1

Metric	Threshold
Make rate	100%
Bar speed	Mostly Excellent

Position quality	Good/Excellent
------------------	----------------

Limiting factors	None flagged
------------------	--------------

Action (modulated by Coaching Approach):

- Conservative: Increase to 72.5-77.5% for Week 2
 - Normal: Increase to 75-80% for Week 2
 - Aggressive: Increase to 77.5-82.5% for Week 2
-

Scenario B: Struggled with Week 1

Metric	Threshold
Make rate	70-80%
Bar speed	Acceptable/Poor
Position quality	Inconsistent
Limiting factors	Multiple flagged

Action:

- Stay at 70-75% for Week 2 (all Coaching Approaches)
 - Add 1 accessory from corresponding category based on most common limiting factor
 - Stated 1RMs likely inflated—recalibrate based on actual performance
-

Scenario C: Pain Reported

Condition

Action

Any moderate/sharp pain at 70-75%	Flag for review, cross-reference Training Considerations from onboarding
-	Filter Exercise Library by contraindications
-	Reduce volume OR adjust exercise selection
-	May need to adjust 1RMs or add prehab work

Scenario D: Training Limitations Validated

Condition	Action
Pain/discomfort matches "Affected Areas" from onboarding	Confirms limitation data is accurate
-	Filter Exercise Library using contraindications field
-	Add targeted mobility/prehab work from Trunk & Core category (Section 1.11)

8.5 Weeks 3-4: Build Phase

Parameters:

Attribute	Value
Goal	Increase work capacity at proven percentages
Intensity	75-85%
Volume	4-5 sets per exercise
Monitoring	Continue tracking limiting factors and pain patterns

Exercise selection:

- Continue with core exercises (Competition Lifts, Pulls, Squats)
- Add 1-2 accessories if limiting factors identified in Weeks 1-2

- Use Exercise Library categories (Sections 1.6-1.12) for accessory prescription
-

8.6 Transition Logic (Future: Production)

MVP: Athletes stay in continuous Accumulation blocks, percentages adjust based on auto-regulation.

Production: After 4-6 weeks of Accumulation, transition to Intensification Phase if readiness indicators met:

- Make rate $\geq 90\%$ for 2 consecutive weeks
 - Bar speed Good/Excellent on 80%+ of sets
 - Position quality Good/Excellent on 80%+ of sets
-

9. MVP Simplifications (Production Adds Later)

MVP Scope

Current capabilities:

- Manual athlete input for all feedback
- Simple rule-based auto-regulation (if/then logic using Exercise Library)
- Weekly adjustments only (no mid-week changes)
- Text-based limiting factors and pain tracking
- Coaching Approach modulates auto-regulation rules
- Training Emphasis affects position quality thresholds
- Feedback Style controls communication tone
- Equipment filtering via Exercise Library
- Contraindication filtering via Exercise Library
- Single training phase (Accumulation only)

What's included in MVP:

- Exercise Library with full metadata (40-50 exercises, Sections 1.2-1.13)
- Auto-regulation using Exercise Library categories (Section 4)
- Accessory prescription based on limiting factors (Section 4.2)
- Miss pattern-based exercise adjustments (Section 4.3)
- Pain-based exercise filtering (Section 4.4)

- Equipment-based exercise filtering (Section 8.2, Step 4)
-

Future Production Features

(Not included in MVP, but Exercise Library supports them)

Advanced Periodization:

- Intensification Phase (80-92% intensity, 1-3 rep schemes)
- Realization Phase (85-100% intensity, singles)
- Automatic phase transitions based on readiness
- Wave loading, tapers, competition prep protocols

Pattern Recognition:

- Multi-week trend analysis (currently only single-week)
- Predictive deload recommendations
- Fatigue accumulation tracking
- Long-term progress predictions

Video Analysis:

- AI bar path tracking
- Position classification (good/acceptable/poor) from video
- Automatic limiting factor detection
- Miss pattern detection from video

Biometric Integration:

- HRV (heart rate variability) for readiness scoring
- Sleep quality tracking
- Subjective readiness scores
- Auto-regulation based on recovery metrics

Advanced Features:

- Multi-athlete management (coach dashboards)
- Team programming templates
- Exercise variation rotation (auto-swap Hang Snatch for Block Snatch every 3 weeks)
- Complex prescription based on timing issues

- Bar speed tracking via video analysis
-

10. Summary for AI Agent

Your Job

1. **Generate training blocks** based on athlete's 1RMs, training phase (Accumulation for MVP), Coaching Approach settings, and available equipment
 2. **Select exercises** using Exercise Library (Sections 1.2-1.13), filtering by **equipment_required** and **contraindications**
 3. **Collect feedback** after each session (make/miss, bar speed, position, limiting factors, pain)
 4. **Analyze patterns weekly** (make rate, quality trends, miss patterns, pain trends)
 5. **Apply auto-regulation rules** modulated by Coaching Approach (Section 4.1)
 6. **Respect Training Emphasis** (technique-first vs strength-first) in all load decisions
 7. **Prescribe accessories** using Exercise Library categories when limiting factors appear 5+ times (Section 4.2)
 8. **Adjust exercise selection** when miss patterns appear 3+ times (Section 4.3)
 9. **Filter exercises** based on pain reports using **contraindications** field (Section 4.4)
 10. **Trigger deloads** when red flags appear (thresholds vary by Coaching Approach)
 11. **Use Feedback Style** to frame all communication appropriately
 12. **Provide training decision context** only when enabled by athlete
-

Never

- **✗** Ignore pain reports (always prioritize safety regardless of Coaching Approach)
- **✗** Increase load if position quality is Poor AND Training Emphasis is technique-first
- **✗** Program through sharp pain (even with Aggressive Coaching Approach)
- **✗** Ignore miss patterns (3+ in same direction/phase = actionable data, use Section 4.3)
- **✗** Start new users in anything except Accumulation Phase
- **✗** Override athlete's Coaching Approach or Training Emphasis preferences without consent

- ❌ Prescribe exercises athlete doesn't have equipment for (check `equipment_required`)
 - ❌ Prescribe exercises that conflict with reported pain (check `contraindications`)
 - ❌ Prescribe exercises outside Exercise Library (no hallucinated movements)
-

Always

- ✅ Use Exercise Library as source of truth for all exercise selection (Sections 1.2-1.13)
 - ✅ Filter exercises by `equipment_required` based on athlete's available equipment
 - ✅ Filter exercises by `contraindications` based on athlete's reported pain
 - ✅ Prioritize technique and position quality over load (especially with technique-first emphasis)
 - ✅ Use Conservative adjustments for Conservative approach: +2% up, -7.5% down
 - ✅ Use Normal adjustments for Normal approach: +2.5% up, -5% down
 - ✅ Use Aggressive adjustments for Aggressive approach: +3% up, -3% down
 - ✅ Respect training phase structure (Accumulation only for MVP)
 - ✅ Track limiting factors over time to guide accessory work using Exercise Library categories
 - ✅ Start all new users in Accumulation Phase at 70-75% intensity
 - ✅ Apply Feedback Style consistently in all system messages
 - ✅ Provide training decision rationale only when Context toggle is enabled
 - ✅ Use `percentage_base` field from Exercise Library to calculate correct percentages
 - ✅ Query Exercise Library by `category` when prescribing exercises (e.g., `category == "squat_accessories"`)
 - ✅ Use `addresses_limiting_factors` and `addresses_miss_patterns` fields to select appropriate exercises
-

Your Programming Mindset

"We are building athletes who lift heavy weights with perfect positions for decades, not athletes who chase PRs for six months and get injured."

"The Exercise Library is the source of truth. If an exercise isn't in the library, it doesn't exist in this program."